



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES



Name: K.B.Devarshan.
Reg.no: 192511426

Course Code:CSA1709
CourseName:Artificial Intelligence

Aim: Exp :4 Write the Python Program for Crypt-Arithmetic Problem

To Write the Python Program for **Crypt-Arithmetic Problem**

Algorithm:

Step 1: Write down the arithmetic problem in full, including any carry-overs.

Step 2: Replace each letter with a unique digit from 0 to 9.

Step 3: Generate all possible combinations of digits for each letter, subject to any constraints ' from the carry-overs and the rules of arithmetic.

Step 4: For each combination of digits, evaluate the arithmetic problem and check if it satisfies the given constraints.

Step 5: If a solution is found, output the values of the letters and the arithmetic problem, and exit the algorithm.

Step 6: If no solution is found, output a message indicating that no solution exists.

Program:

```
import itertools

def solve_cryptarithmic(word1, word2, result):
    letters = set(word1 + word2 + result)

    if len(letters) > 10:
        print("No solution exists.")
        return

    letters = list(letters)
    first_letters = {word1[0], word2[0], result[0]}

    for perm in itertools.permutations(range(10), len(letters)):
        mapping = dict(zip(letters, perm))

        if any(mapping[letter] == 0 for letter in first_letters):
```

```
continue
```

```
num1 = int("".join(str(mapping[l]) for l in word1))  
num2 = int("".join(str(mapping[l]) for l in word2))  
num_result = int("".join(str(mapping[l]) for l in result))
```

```
if num1 + num2 == num_result:  
    print("Solution found:")  
    for letter in sorted(mapping.keys()):  
        print(letter, "=", mapping[letter])  
    print(num1, "+", num2, "=", num_result)  
    return
```

```
print("No solution exists.")
```

```
if __name__ == '__main__':  
    solve_cryptarithmic("SEND", "MORE", "MONEY")
```

OUTPUT:

```
Solution found:  
D = 7  
E = 5  
M = 1  
N = 6  
O = 0  
R = 8  
S = 9  
Y = 2  
9567 + 1085 = 10652
```

Result: Thus, the program was executed and found that the output was found to be correct.